Pure Mathematics with Mechanics GCE A Level 2026 - Paper 1

Duration: 1h30 | Cameroon GCE Board

Multiple choice paper (0765 Pure Maths With Mechanics 1, June 2026, Advanced Level, one and a half hours, paper code GCE26/0765/1/P52). Answer ALL questions; each question has FOUR suggested answers A, B, C and D and only one answer must be marked (if more than one answer is marked the score for that question is zero). Formulae booklets and calculators are allowed. Section A: Pure Mathematics (questions 1 to 35); Section B: Mechanics (questions 36 to 50). Mobile phones are NOT allowed in the examination room.

SECTION A: PURE MATHEMATICS

If $\frac{M}{x-1}+\frac{N}{x+2}\equiv\frac{2x-1}{(x-1)(x+2)}$, then

a) $M=\frac{1}{3},\ N=-\frac{5}{3}$
b) $M=\frac{1}{3},\ N=\frac{5}{3}$
c) $M=-3,\ N=3$
d) $M=3,\ N=-3$

2. Given the function $f:\mathbb{R}\to\mathbb{R}$, where $f(x)=\frac{x+1}{x+2},\ x\neq-2$, the range of f does not include the element

a) 1
b) 2
c) -2
d) -1

3. The values of x that satisfy the equation $2^{2x}-5\left(2^{x+1}\right)+2^4=0$ are

a) $x=3$ or $x=1$
b) $x=-3$ or $x=1$
c) $x=3$ or $x=-1$
d) $x=-3$ or $x=-1$

4. The solution of the differential equation $y\frac{dy}{dx}=3x$, given that $y=1$ when $x=1$, is

a) $y^2=3x^2-2$
b) $y^2=3x^2+1$
c) $y^2=3x^2-1$
d) $y^2=3x^2+2$

5. In the set $A=\{0,1,2,3,4,5,6,7\}$, a relation R is defined by $x\,\mathrm{R}\,y\Leftrightarrow x$ and y have the same remainder when divided by 3. The equivalence class of 3, i.e. $[3]$, is

a) $[3] = \{3, 6\}$
b) $[3] = \{1, 4, 7\}$
c) $[3] = \{2, 5\}$
d) $[3] = \{0, 3, 6\}$

6. The line segment PQ, where P is the point $(2, 4)$ and Q is the point $(8, 10)$, is the diameter of a circle. The equation of the circle is

a) $(x-2)(x-8) + (y-4)(y-10) = 0$
b) $(x+2)(x+8) - (y+4)(y+10) = 0$
c) $(x-2)(x-4) + (y-8)(y-10) = 0$
d) $(x+2)(x+4) - (y+8)(y+10) = 0$

7. When $f(x) = 2x^3 + x^2 - 13x + 6$ is divided by $2x + 1$, the remainder is

a) 0
b) -0.5
c) 12.5
d) 0.5

8. The range of values of x for which $|x - 4| \geq 2$ is

a) $x \geq 2$
b) $x \leq 6$
c) $2 \leq x \leq 6$
d) $x \leq 2$ or $x \geq 6$

9. Consider the two statements below.

p: John is sick.

q: He goes to the hospital.

The contrapositive of $p \Rightarrow q$ is

a) $q \Rightarrow p$
b) $\sim q \Rightarrow p$
c) $\sim p \Rightarrow \sim q$
d) $\sim q \Rightarrow \sim p$

10. Given that $f(x) = \begin{vmatrix} 0 & 2 & x^2 \\ 0 & 1 & x^2 \\ 1 & 3x & 2x^2 \end{vmatrix}$, $f'(x) =$

a) $2x$
b) $6x$
c) $2x^2$
d) $6x^2$

11. Given that f is a periodic function of period 4 and that $f(x)=\begin{cases}9-x^2, & 0

a) 56
b) -352

c) 0
d) 8

12. The volume generated when the area of the finite region enclosed by the x-axis and the curve $y = 1 - x^2$ is rotated completely about the x-axis is

a) $\pi \int_0^1 (1 - x^2)^2 \, dx$
b) $2\pi \int_{-1}^1 (1 - x^2)^2 \, dx$
c) $\pi \int_{-1}^1 (1 - x^2)^2 \, dx$
d) $2\pi \int_{-1}^1 (1 - x^2) \, dx$

13. If a root of an equation $f(x) = 0$ lies between 0 and 1, then

Options as printed: A) $f(0)f(1) \leq 0$ B) $f(0)f(1) > 0$ C) $f(0)f(1) \geq 0$ D) $f(0)f(1) < 0$

14. The vectors $\mathbf{p}$ and $\mathbf{q}$ are such that $|\mathbf{p}| = 3$, $|\mathbf{q}| = 5$ and $\mathbf{p} \cdot \mathbf{q} = -14$. The value of $|\mathbf{p} - \mathbf{q}| =$

a) $\sqrt{44}$
b) 44
c) $\sqrt{62}$
d) 62

15. The sum of the first n terms of a sequence is given by $S_n = 2n^2 + n$. The nth term of the sequence is

a) $2n^2 + 1$
b) $4n - 1$
c) $4n + 1$
d) $2n^2 - 1$

16. The expansion of $(3 + 5x)^{-2}$ is valid when

a) $-\dfrac{3}{5} \leq x \leq \dfrac{3}{5}$
b) $-\dfrac{3}{5} < x < \dfrac{3}{5}$ c) $-\dfrac{3}{5} \leq x < \dfrac{3}{5}$ d) $-\dfrac{3}{5} \leq x < \dfrac{3}{5}$

17. The Cartesian equation of the curve with parametric equation $x = 2t, \; y = \dfrac{2}{t}$, where t is a parameter is

a) $y = x$
b) $y = \dfrac{1}{x}$
c) $y = \dfrac{4}{x}$
d) $y = \dfrac{4}{x}$

18. $\lim\limits_{x \to 1} \left(\dfrac{x^2 - 1}{x - 1} \right) =$

a) 2
b) -2

c) 0
d) ∞

19. $\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\cdots=$

a) $\frac{1}{4}$
b) $\frac{1}{3}$
c) 2
d) 1

20. The asymptotes of the curve $y=\dfrac{1}{x^2-5x+6}$ are

a) $x=-2,\ x=-3,\ y=0$
b) $x=2,\ x=3,\ y=0$
c) $x=-2,\ x=-3,\ y=1$
d) $x=2,\ x=3,\ y=1$

21. The values of y corresponding to the values of x are given in the table below.

x	**6**	**9**	**12**	**15**	**18**	**21**
y	0.3	0.8	a	2.1	3.0	4.3

Using the trapezium rule, $\int_6^{21} y\,dx \approx 28.8$. The value of a is

a) 1.4
b) 2.8
c) 1.8
d) 2.4

22. The gradient of the implicit function $2x+xy=5$ at the point $(1,1)$ is

a) -3
b) 3
c) -2
d) 2

23. $\sin 35^\circ+\sin 25^\circ=$

a) $2\cos 5^\circ$
b) $\cos 5^\circ$
c) $4\cos 5^\circ$
d) $\frac{1}{2}\cos 5^\circ$

24. Given that the plane $\mu x-2y+5z=6$ and the line $\mathbf{r}=2\mathbf{i}-3\mathbf{j}+\mathbf{k}+s(\mathbf{i}-2\mathbf{j}+3\mathbf{k})$ are parallel, find the value of the constant μ.

a) -19
b) 19

c) $-\frac{11}{2}$
d) $\frac{11}{2}$

25. $\int \frac{x}{2+3x^2}\,dx =$

a) $\ln(2+3x^2)+K$
b) $6\ln(2+3x^2)+K$
c) $\frac{1}{6}\ln(2+3x^2)+K$
d) $\frac{1}{6}\ln\left(\frac{x}{2+3x^2}\right)+K$

26. The general solution of the equation $\cos\left(2x+\frac{\pi}{3}\right)=\frac{\sqrt{3}}{2}$ is

a) $2x = 2n\pi \pm \frac{\pi}{6} - \frac{\pi}{3}$
b) $2x = 2n\pi + \frac{\pi}{6} - \frac{\pi}{3}$
c) $2x = 2n\pi$
d) $2x = 2n\pi - \frac{2\pi}{3}$

27. The Cartesian equation of the line $\mathbf{r} = \mathbf{i} - \mathbf{j} + 6\mathbf{k} + \mu(-5\mathbf{i} + \mathbf{j} - 4\mathbf{k})$ is

a) $\frac{x-5}{1} = \frac{y-1}{-1} = \frac{z-4}{6}$
b) $\frac{x+5}{1} = \frac{y-1}{-1} = \frac{z+4}{6}$
c) $\frac{x-1}{-5} = \frac{y+1}{1} = \frac{z-6}{-4}$
d) $\frac{x+1}{-5} = \frac{y-1}{1} = \frac{z+6}{-4}$

28. If the sequence u_n where $n \in \mathbb{N}$ is strictly monotonic increasing then

a) $u_{n+1} b) $u_{n+1} \leq u_n$
c) $u_{n+1} > u_n$
d) $u_{n+1} \geq u_n$

29. A bijective function is that which is

a) reflexive, symmetric and transitive
b) injective and surjective
c) injective, surjective and anti-symmetric
d) Monotonic and bounded

30. The complex number $z = 1 + i$ can be expressed in exponential form as

a) $e^{\frac{\pi}{4}i}$
b) $e^{\frac{-\pi}{4}i}$
c) $\sqrt{2}\,e^{\frac{\pi}{4}i}$
d) $2e^{\frac{\pi}{4}i}$

31. $\dfrac{\cos 5\theta + i\sin 5\theta}{(\cos 2\theta + i\sin 2\theta)^{-3}} =$

a) $\cos 7\theta + i\sin 7\theta$
b) $\cos\theta - i\sin\theta$
c) $\cos\theta + i\sin\theta$
d) $\cos 11\theta + i\sin 11\theta$

32. Suppose $f(0.6) = 0.7$ and $f'(0.6) = 10.6$. If $x_1 = 0.6$ is the first approximation to the root of $f(x) = 0$ then using the Newton-Raphson method, a second approximate root x_2 is

a) $x_2 = 0.6 + \dfrac{0.7}{10.6}$
b) $x_2 = 0.6 + \dfrac{10.6}{0.7}$
c) $x_2 = 0.6 - \dfrac{0.7}{10.6}$
d) $x_2 = 0.6 - \dfrac{10.6}{0.7}$

33. Given the matrix $\mathbf{A}$, where $\mathbf{A} = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 3 & 1 \\ 0 & -1 & 4 \end{pmatrix}$, the determinant of $\mathbf{A}^{\mathrm{T}}$, where $\mathbf{A}^{\mathrm{T}}$ is the transpose of $\mathbf{A}$, is

a) 12
b) 13
c) 3
d) 21

34. The image of the line $y = 3x$ under the transformation matrix $\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$ is the line

a) $y = 15x$
b) $y = 5x$
c) $y = \dfrac{5}{10}x$
d) $y = 2x$

35.

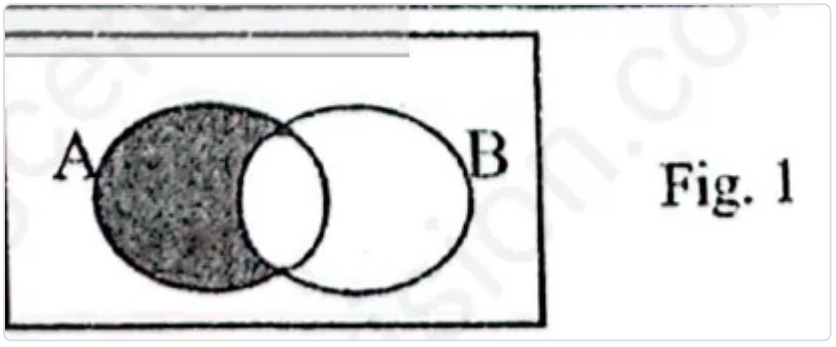


Fig. 1

The shaded region, in Fig. 1, represents

a) $A' \cap (A \cap B)$
b) $A' \cap B$
c) $A \cap B'$
d) $B' \cap (A \cap B)$

SECTION B: MECHANICS

The position vector of a particle is $\mathbf{r} = [(3\sin 3t)\mathbf{i} + (2\cos 2t)\mathbf{j}]$ m at time t s. The initial velocity of the particle is

a) $9\mathbf{i}\ \text{m s}^{-1}$
b) $(9\mathbf{i} + 4\mathbf{j})\ \text{m s}^{-1}$
c) $(9\mathbf{i} - 4\mathbf{j})\ \text{m s}^{-1}$
d) $9\ \text{m s}^{-1}$

37. When an elastic string is stretched to a length of 1.8 m, the tension in the string is 14 N. Given that the modulus of elasticity of the string is 28 N, the natural length of the string is

a) 1.6 m
b) 0.6 m
c) 0.12 m
d) 1.2 m

38. The speeds of a car during 8 seconds of braking are given in the table below.

Time, t s	**0**	**2**	**4**	**6**	**8**
Speed, $v\ \text{m s}^{-1}$	30	27	22	14	0

Using the trapezium rule, the distance moved by the car in coming to rest is

a) 156 m
b) 78 m
c) 93 m
d) 186 m

39. Fig. 2 shows the forces acting on a uniform ladder AB, of length 12 m, resting in limiting equilibrium. A man M ascends the ladder.

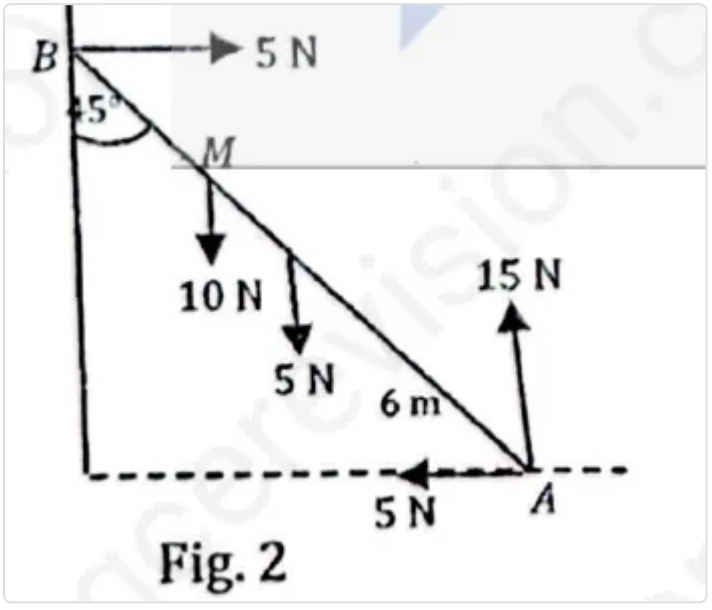


Fig. 2

The ladder will begin to slip when M has ascended up the ladder a distance

Options as printed: A) 9 m B) 6 m C) 3 m D) 4 m

40. A girl on a ship, whose velocity is $(4\mathbf{i} - 7\mathbf{j})\ \mathrm{m\,s^{-1}}$ is watching a boat whose velocity is $(-8\mathbf{i} - 2\mathbf{j})\ \mathrm{m\,s^{-1}}$. The speed of the boat as it appears to the girl on the ship is

a) $(-12\mathbf{i} + 5\mathbf{j})\ \mathrm{m\,s^{-1}}$
b) $(12\mathbf{i} - 5\mathbf{j})\ \mathrm{m\,s^{-1}}$
c) $13\ \mathrm{m\,s^{-1}}$
d) $(\sqrt{68} - \sqrt{65})\ \mathrm{m\,s^{-1}}$

41. The speed v of a particle moving in a straight line from rest t s after passing through a fixed point O is $v = (6t^2 - 5t + 3)\ \mathrm{m\,s^{-1}}$. The distance of the particle from O when $t = 4$ is

a) 43 m
b) 100 m
c) 76 m
d) 46 m

42. The work done by a force $\mathbf{F}$, where $\mathbf{F} = (\mathbf{i} + 2\mathbf{j} + 3\mathbf{k})$ N, which moves its point of application from a point P with position vector $\mathbf{r}_P = (3\mathbf{i} + 4\mathbf{j} + 5\mathbf{k})$ m to another point Q whose position vector is $\mathbf{r}_Q = (7\mathbf{i} + 6\mathbf{j} + 5\mathbf{k})$ m is

a) -60 J
b) 60 J
c) -8 J
d) 8 J

43. A car of mass $1{,}200$ kg moving with speed $12\ \mathrm{m\,s^{-1}}$ has its engine working at 54 kW up a road inclined at an angle $\sin^{-1}\left(\frac{1}{5}\right)$ to the horizontal. The non-gravitational resistance to the motion of the car is 300 N. Taking g as $10\ \mathrm{m\,s^{-2}}$, the acceleration of the car is

a) $2.5\ \mathrm{m\,s^{-2}}$
b) $1.5\ \mathrm{m\,s^{-2}}$
c) $3.5\ \mathrm{m\,s^{-2}}$
d) $1.2\ \mathrm{m\,s^{-2}}$

44. Three particles of weight 7 N, 9 N and 4 N have position vector $(4\mathbf{i} + \mathbf{j})$ m, $3\mathbf{j}$ m and $(3\mathbf{i} + 4\mathbf{j})$ m respectively. The position vector of their centre of mass is

a) $\left(\frac{7}{3}\mathbf{i} + \frac{8}{3}\mathbf{j}\right)$ m
b) $\left(2\mathbf{i} + \frac{5}{2}\mathbf{j}\right)$ m
c) $(40\mathbf{i} + 50\mathbf{j})$ m
d) $\left(\frac{7}{20}\mathbf{i} + \frac{8}{20}\mathbf{j}\right)$ m

45. Fig. 3 shows two particles P and Q hanging freely and connected by a light inextensible string passing over a fixed smooth pulley.

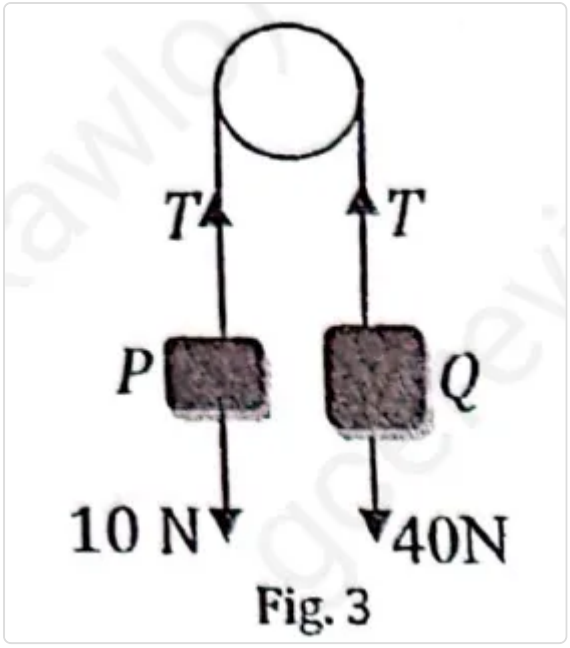


Fig. 3

The system is released from rest. Taking g as $10\ \mathrm{m\,s^{-1}}$, the acceleration of P and the tension in the string are

a) $6\ \mathrm{m\,s^{-2}}$ and 8 N
b) $10\ \mathrm{m\,s^{-2}}$ and 8 N
c) $10\ \mathrm{m\,s^{-2}}$ and 16 N
d) $6\ \mathrm{m\,s^{-2}}$ and 16 N

46. A particle of mass 4 kg which is moving in horizontal straight line with velocity $15\ \mathrm{m\,s^{-1}}$ strikes a vertical wall. If the coefficient of restitution between the particle and the wall is $\frac{2}{3}$ then the magnitude of the impulse experienced by the particle is

a) 100 N s
b) 40 N s
c) 60 N s
d) 20 N s

47. A particle is projected from a point O with initial speed $(15\sqrt{3}\,\mathbf{i} + 15\mathbf{j})\ \mathrm{m\,s^{-1}}$. Taking [$g$] as $10\ \mathrm{m\,s^{-2}}$ its position vector after 2 seconds is

a) $(15\sqrt{3}\,\mathbf{i} - 5\mathbf{j})$ m
b) $(30\sqrt{3}\,\mathbf{i} + 10\mathbf{j})$ m
c) $(10\mathbf{i} + 30\sqrt{3}\,\mathbf{j})$ m
d) $(-5\mathbf{i} + 15\sqrt{3}\,\mathbf{j})$ m

48. One end of an inelastic string is fixed to a point O and the other end is attached to a particle of mass 4 kg. The particle is kept in a horizontal circle of radius 8 m. Given that the center of the circle is 2.5 m vertically below O and the tension in the string is 128 N, the angular speed of the particle is

Options as printed: A) $16\ \mathrm{m\,s^{-1}}$ B) $16\ \mathrm{rad\,s^{-1}}$ C) $2\ \mathrm{rad\,s^{-1}}$ D) $2\ \mathrm{m\,s^{-1}}$

49. A particle of mass m moving along a straight line Ox is subjected to a force $16mx$ acting towards the origin O. Given that the particle starts from rest when $x = 5$ m, the speed of the particle as it passes through the O is

a) $6\ \mathrm{m\,s^{-1}}$
b) $\frac{16}{5}\ \mathrm{m\,s^{-1}}$

c) $20\ \mathrm{m\,s^{-1}}$
d) $10\sqrt{2}\ \mathrm{m\,s^{-1}}$

50. If it is known that the events A and B are collectively exhaustive and independent, the true statement is

a) $P(A) + P(B) = P(A) \times P(B)$
b) $P(A) + P(B) = 1 - P(A) \times P(B)$
c) $P(A) + P(B) = 1 + P(A) \times P(B)$
d) $P(A) + P(B) = 1$

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